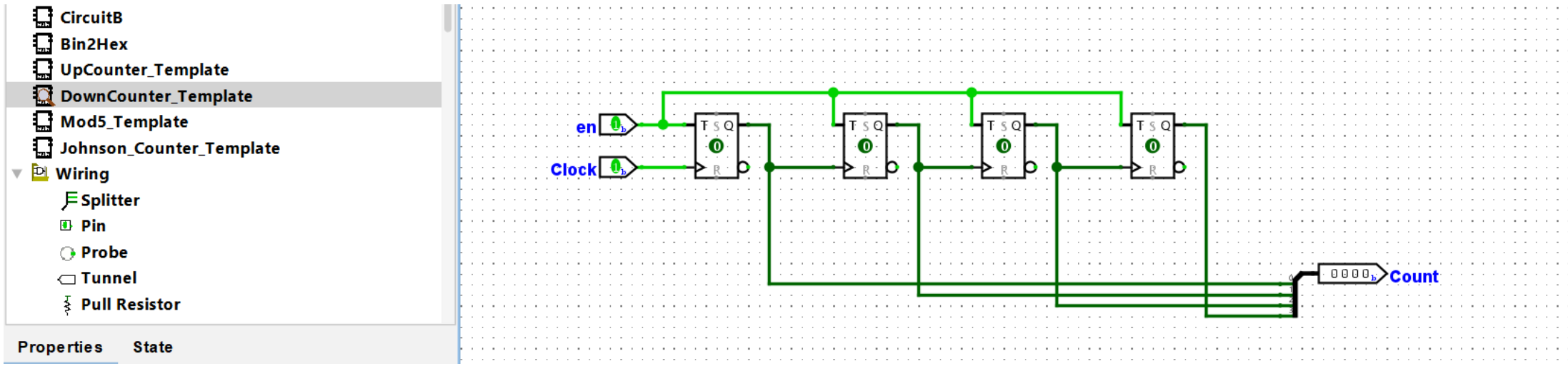
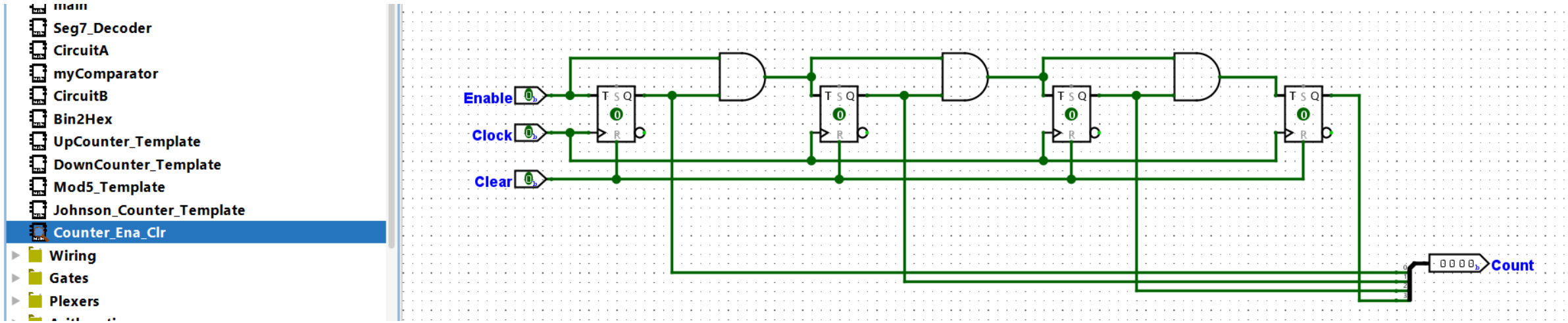
Lab 6

Micah Angle

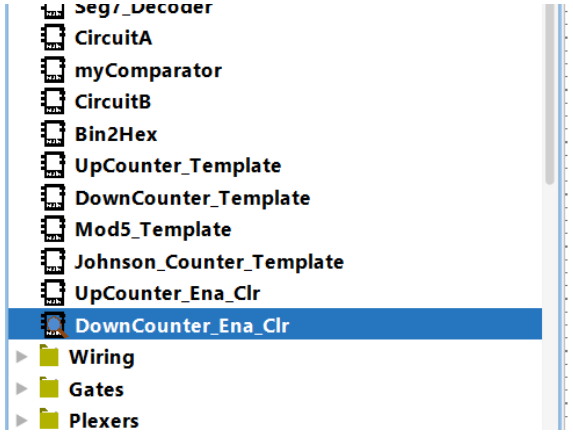
Down Counter

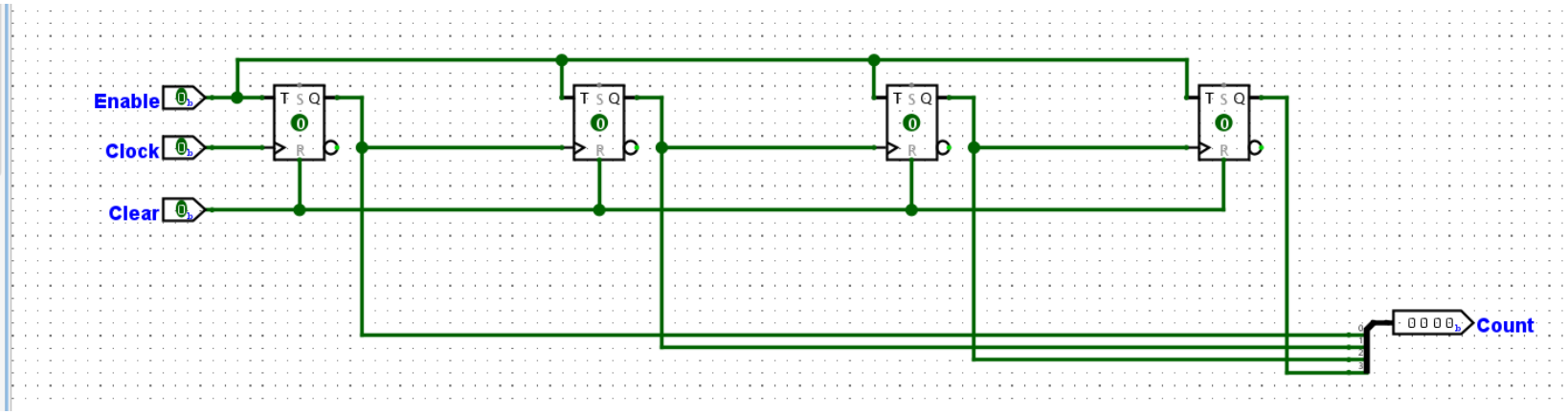

CircuitB
Bin2Hex
UpCounter_Template
DownCounter_Template
Mod5_Template
Johnson_Counter_Template
Wiring
Splitter
Pin
Probe
Tunnel
Pull Resistor
Properties
State
en
Clock
T S Q
R
Count

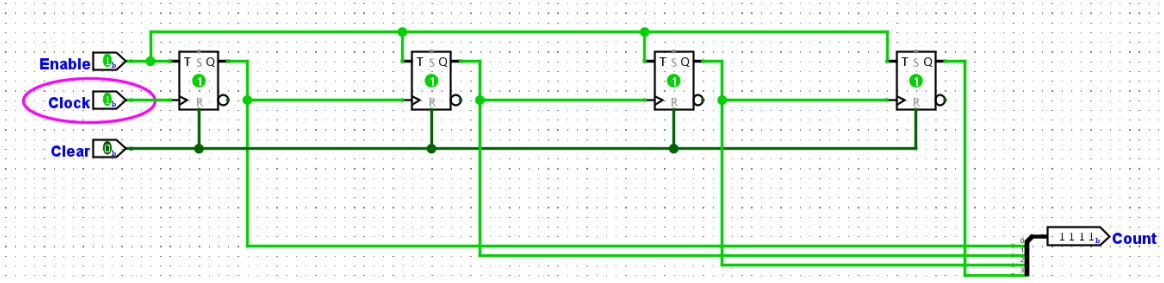
Up Counter w/ Reset

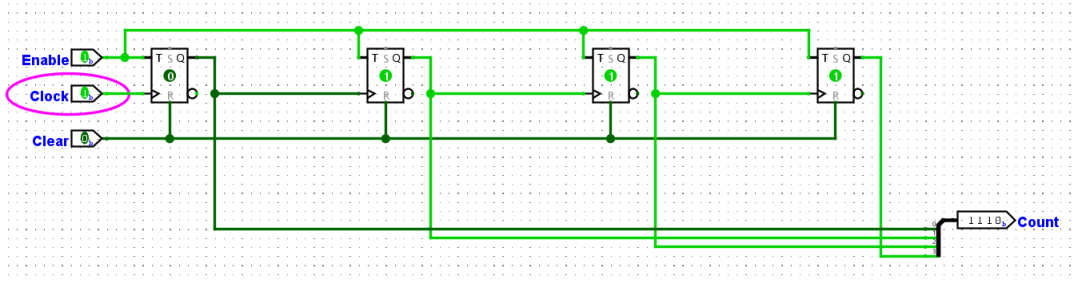

main
Seg7_Decoder
CircuitA
myComparator
CircuitB
Bin2Hex
UpCounter_Template
DownCounter_Template
Mod5_Template
Johnson_Counter_Template
Counter_Ena_Clr
Wiring
Gates
Plexers
Enable
Clock
Clear
T S Q
R
Count

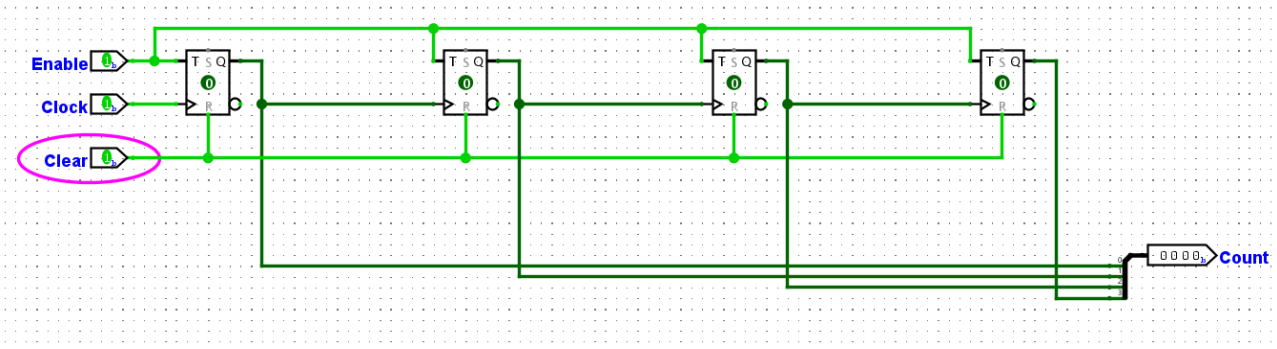
Down Counter w/ Reset


Seg7_Decoder
CircuitA
myComparator
CircuitB
Bin2Hex
UpCounter_Template
DownCounter_Template
Mod5_Template
Johnson_Counter_Template
UpCounter_Ena_Clr
DownCounter_Ena_Clr
Wiring
Gates
Plexers

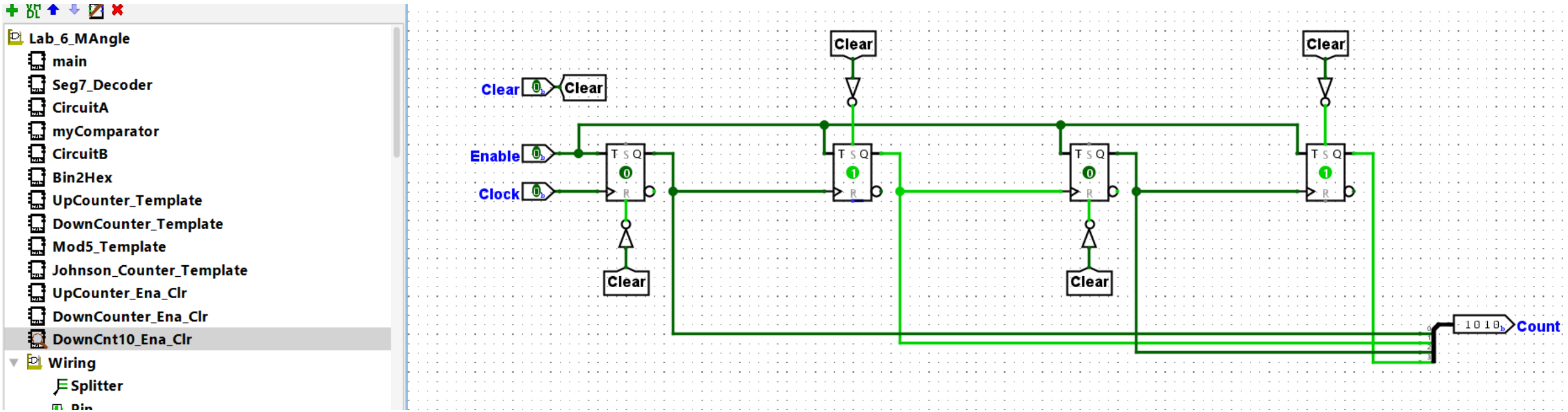

Enable
Clock
Clear
T S Q
R
0000b
Count


Enable
Clock
Clear
T S Q
R
1111b
Count

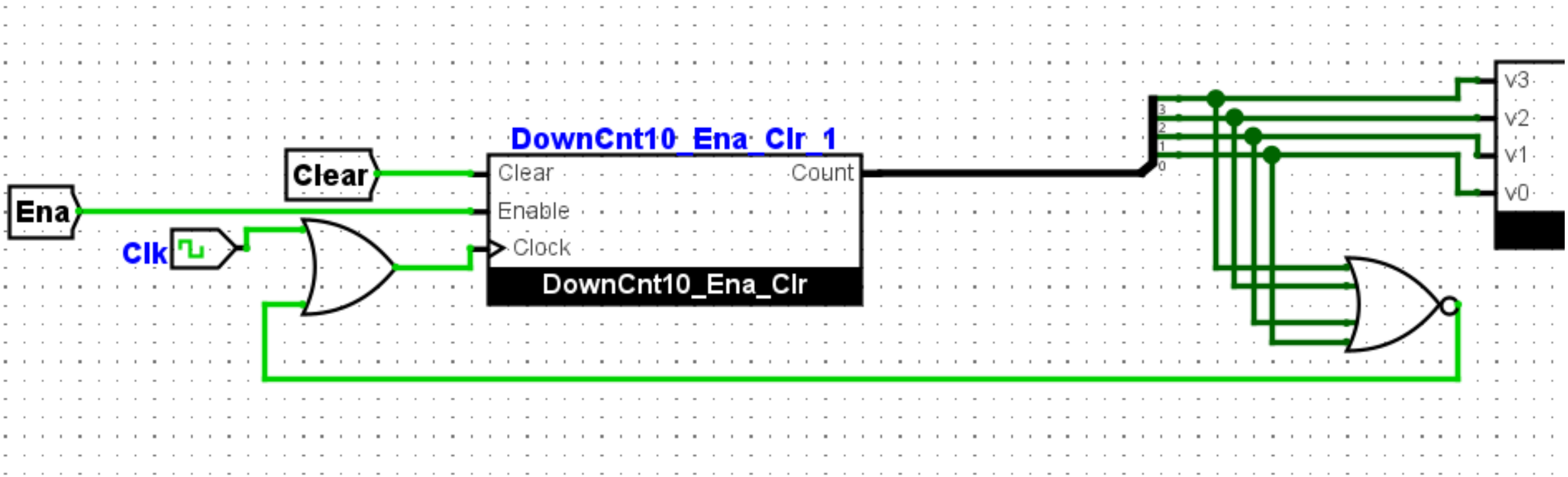

Enable
Clock
Clear
T S Q
R
1110b
Count


Enable
Clock
Clear
T S Q
R
0000b
Count

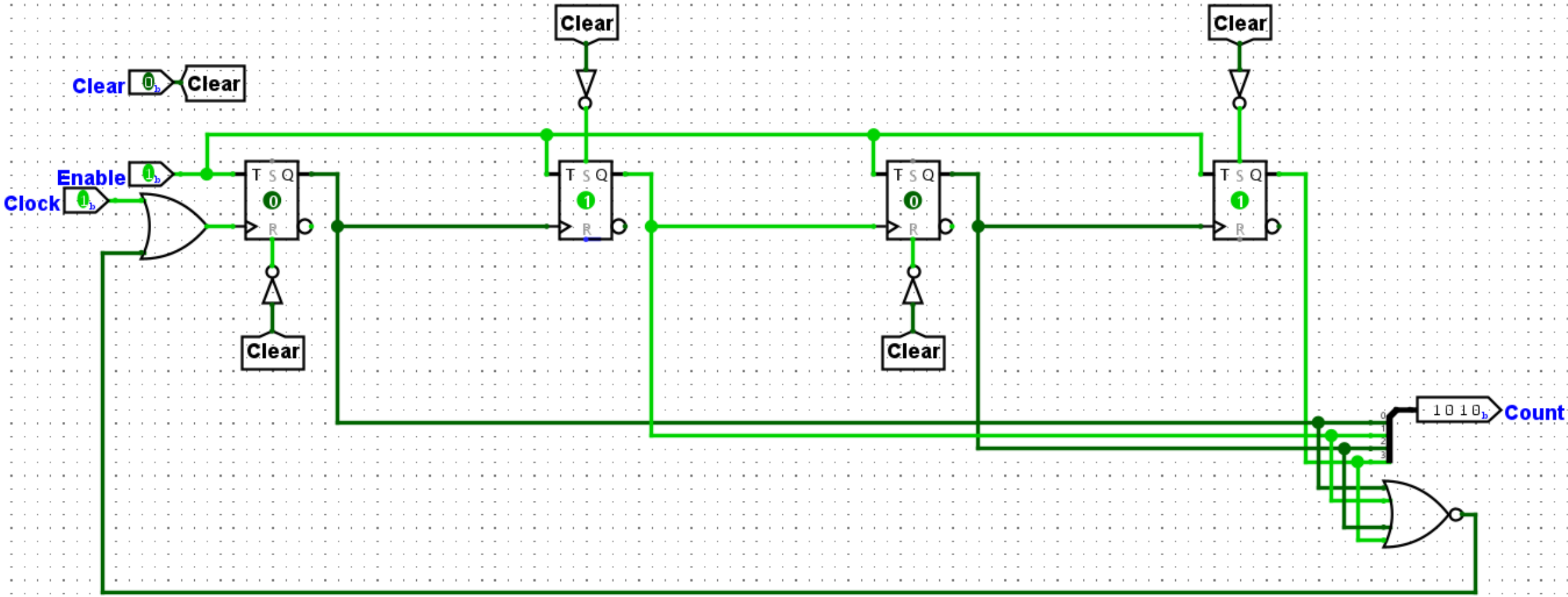
Down Counter from 10 w/ Reset


Lab_6_MAngle
main
Seg7_Decoder
CircuitA
myComparator
CircuitB
Bin2Hex
UpCounter_Template
DownCounter_Template
Mod5_Template
Johnson_Counter_Template
UpCounter_Ena_Clr
DownCounter_Ena_Clr
DownCnt10_Ena_Clr
Wiring
Splitter
Clear
Clear
Enable
Clock
Clear
Clear
Clear
Clear
1010b
Count

Latch at 0


DownCnt10_Ena_Clr_1
Clear
Ena
Clk
Clear
Enable
Clock
Count
DownCnt10_Ena_Clr
3
2
1
0
V3
V2
V1
V0

Incorporated into Counter

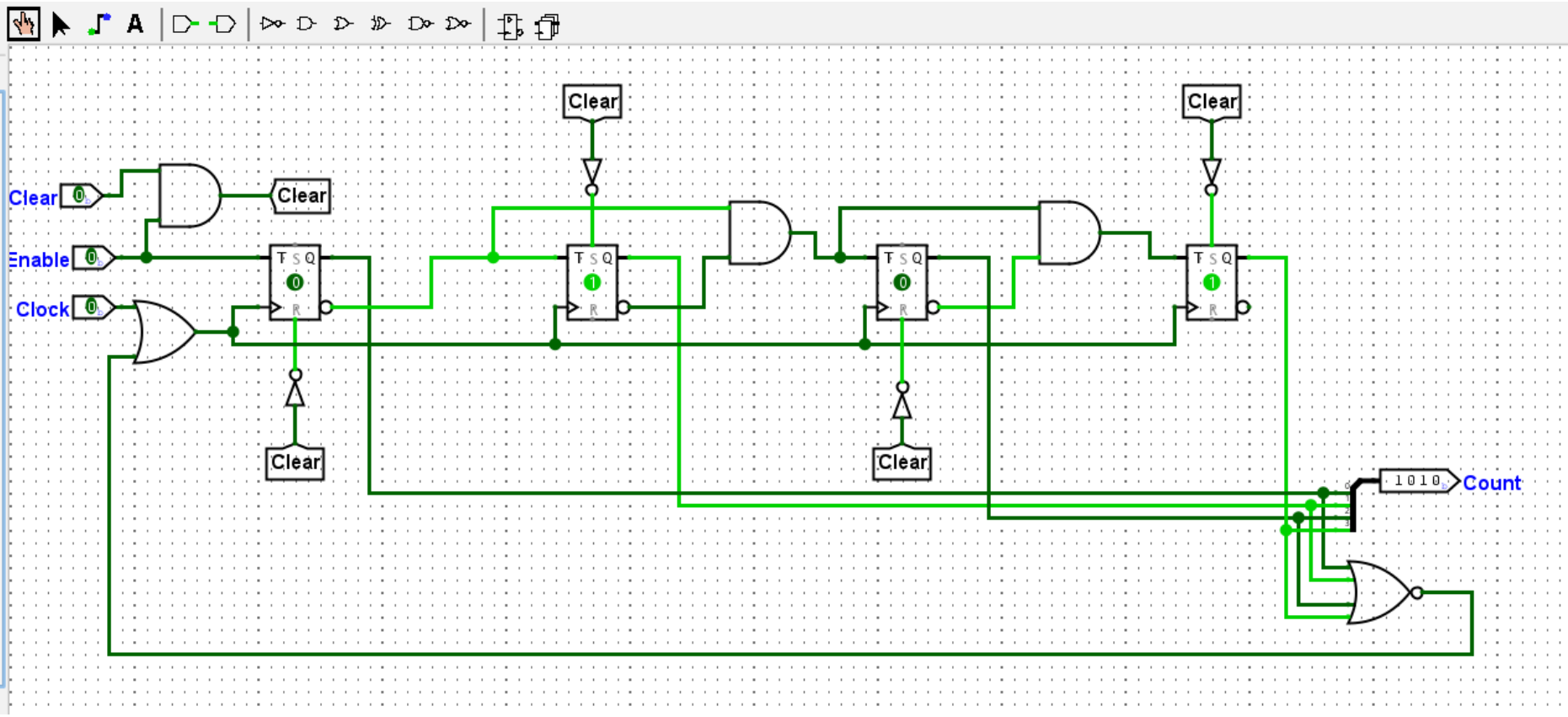

Clear
Clear
Clear
Clear
Enable
Clock
T S Q
0
R
T S Q
1
R
T S Q
0
R
T S Q
1
R
Clear
Clear
1 0 1 0b
Count

Synchronous Down Counter with 0 Latch and 10 Reset

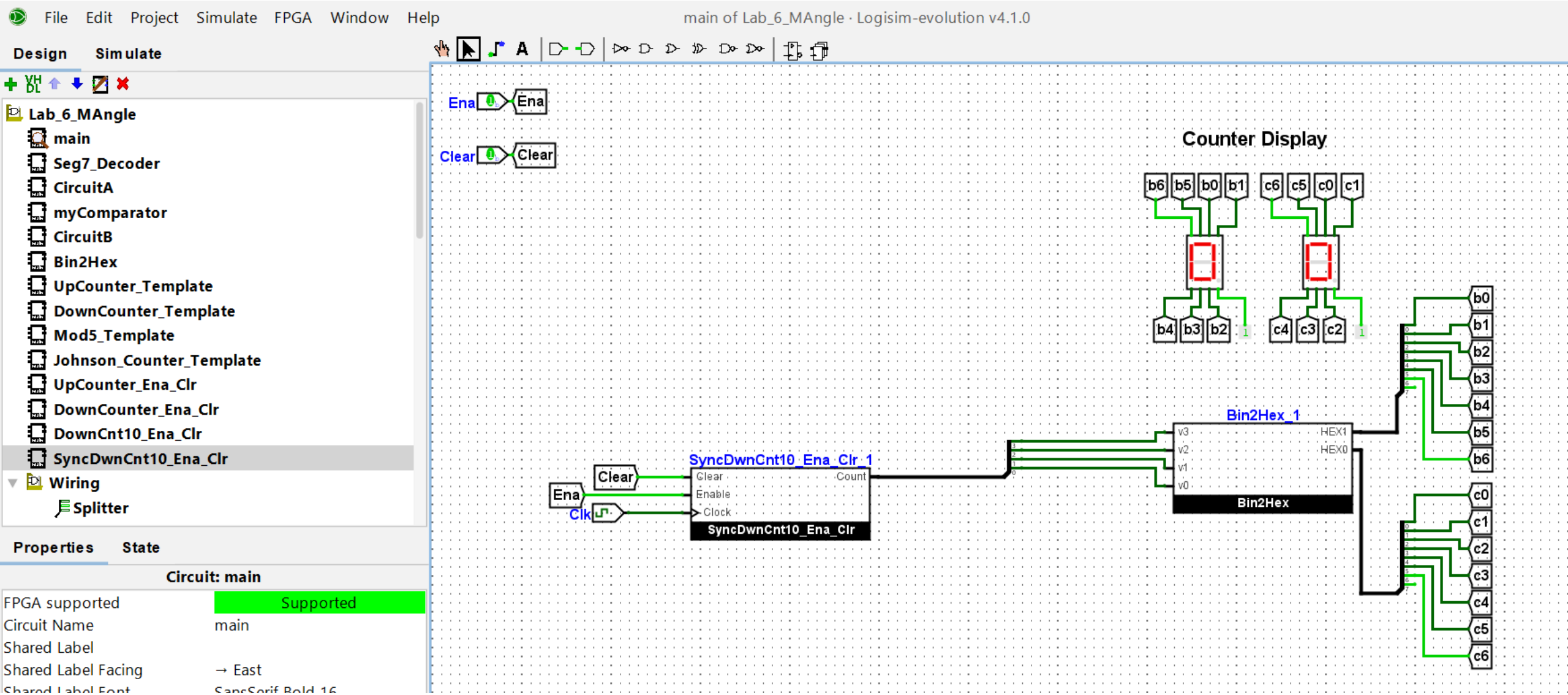
FPGA had issues counting down from 10 asynchronously and counted starting from 8 despite Logisim counting down from 10. This circuit was designed to test if synchronous timing worked better

Design Simulate

- Lab_6_MAngle
  - main
  - Seg7_Decoder
  - CircuitA
  - myComparator
  - CircuitB
  - Bin2Hex
  - UpCounter_Template
  - DownCounter_Template
  - Mod5_Template
  - Johnson_Counter_Template
  - UpCounter_Ena_Clr
  - DownCounter_Ena_Clr
  - DownCnt10_Ena_Clr
  - SyncDwnCnt10_Ena_Clr
  - Wiring
  - Gates



Final Circuit


File
Edit
Project
Simulate
FPGA
Window
Help
main of Lab_6_MAngle · Logisim-evolution v4.1.0
Design
Simulate
Lab_6_MAngle
main
Seg7_Decoder
CircuitA
myComparator
CircuitB
Bin2Hex
UpCounter_Template
DownCounter_Template
Mod5_Template
Johnson_Counter_Template
UpCounter_Ena_Clr
DownCounter_Ena_Clr
DownCnt10_Ena_Clr
SyncDwnCnt10_Ena_Clr
Wiring
Splitter
Properties
State
Circuit: main
FPGA supported
Supported
Circuit Name
main
Shared Label
Shared Label Facing
→ East
Ena
Clear
Counter Display
b6 b5 b0 b1
c6 c5 c0 c1
b4 b3 b2
c4 c3 c2
SyncDwnCnt10_Ena_Clr_1
Clear
Ena
Clk
Clear
Enable
Clock
Count
SyncDwnCnt10_Ena_Clr
Bin2Hex_1
v3
v2
v1
v0
HEX1
HEX0
Bin2Hex
b0
b1
b2
b3
b4
b5
b6
c0
c1
c2
c3
c4
c5
c6